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Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	PMAT1/N08

Set 8

Time: 25 minutes | Paper Reference **PMAT1/N08**

Mathematics

Level 1
Section A (Non–Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
14 /14

My signature confirms that I will not discuss the content of the test with anyone.

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Instructions

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- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**
- Take the value of π to be 3.14

Information

- The total mark for this section is 14.
- The marks for **each** question are shown in brackets.
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1

$7 \times 13 = 91$

(a) What is $91 \div 7$?

(1)

Inverse operations: Multiplication and division are opposites:

- *7 x 13 = 91 --> 91 ÷ 7 = 13 [A1]*

13 [A1]

(b) Work out $4 + 5 \times 3$

(1)

Order of operations (BIDMAS): multiplication before addition:

- *Step 1: Multiply: 5 × 3 = 15*
- *Step 2: Add: 4 + 15 = 19 [B1]*

19 [B1]

(c) Use a reverse calculation to show a check of your answer for part (b).

(1)

Reverse calculation check (reverse addition with subtraction):

- *19 − 15 = 4 (or 19 − 4 = 15; 15 ÷ 3 = 5) [C1]*

(Confirms the original calculation is correct)

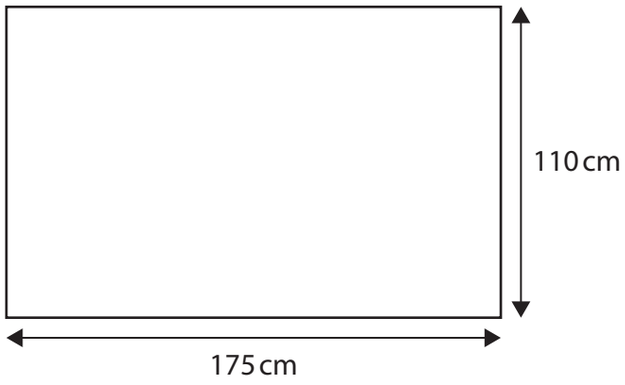
19 − 15 = 4 [C1]

(Total for Question 1 is 3 marks)

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2 Esme is putting copper tape around a flower bed in her garden.

The flower bed is rectangular in shape.
She has this sketch of the flower bed.



Esme wants enough tape to go around all the edges of the flower bed 3 times.

Work out the length of tape Esme needs.
Give your answer in cm.

(3)

Step 1: Calculate the perimeter of the rectangular flower bed:

- *Perimeter = 2 × (Length + Width) = 2 × (175 + 110)*
- ***Perimeter = 2 × 285 = 570 cm [M1]***

Step 2: Calculate tape needed for 3 times around the bed:

- ***Total tape length = 3 × 570 cm = 1710 cm [M1, A1]***

Alternative check: (175 × 6) + (110 × 6) = 1050 + 660 = 1710 cm (Confirmed)

1710 *[A1]* cm

(Total for Question 2 is 3 marks)

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3 Jeff is looking for car insurance.
He must tell the insurance company the mean number of miles he drives his car each year.

Jeff bought his car 5 years ago.

Jeff knows when he bought the car the mileage was 27 563
Today the mileage is 72 889

Work out the mean mileage each year for the last 5 years.

(4)

Step 1: Calculate total miles driven over the 5 years:

- *Total miles = 72 889 − 27 563*
- **Total miles = 45 326 miles [M1]**

Step 2: Calculate the mean mileage per year (divide by 5):

- *Mean = 45 326 ÷ 5 [M1]*
- *45 000 ÷ 5 = 9000; 326 ÷ 5 = 65.2*
- **Mean = 9065.2 miles [M1, A1]**

Verification check:

- *9065.2 × 5 = 45 326; 45 326 + 27 563 = 72 889 (Confirmed)*

9065.2 *[A1]* miles

(Total for Question 3 is 4 marks)

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4 Jess buys sweets for her shop at a wholesalers.

On Tuesday Jess sees sweets with a price of £147.65
The price does not include VAT.

Jess wants to estimate the total cost including VAT.
VAT is 20%

Work out an estimate for the total cost of these sweets.
You **must** show your working.

(4)

Step 1: Round price to a sensible estimate for mental calculation:

- *Estimated sweet price = £150 (rounded to nearest £10) [M1]*

Step 2: Calculate 20% VAT on the estimated price:

- *10% of £150 = £150 ÷ 10 = £15*
- *20% of £150 = £15 × 2 = £30 VAT [M1]*

Step 3: Calculate total estimated cost including VAT:

- *Total estimate = £150 + £30 = £180 [M1, A1]*

Alternative check (using 1.20 multiplier):

- *£150 × 1.20 = £180 (Confirmed)*

£ *180* *[A1]*

(Total for Question 4 is 4 marks)

TOTAL FOR SECTION A IS 14 MARKS

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Set 8

Time: 1 hour 30 minutes | Paper Reference **PMAT1/C08**

Mathematics

Level 1
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
42 /42

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- **Calculators may be used.**
- If your calculator does not have a π button take the value of π to be 3.14

Information

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- The total mark for this paper is 56.
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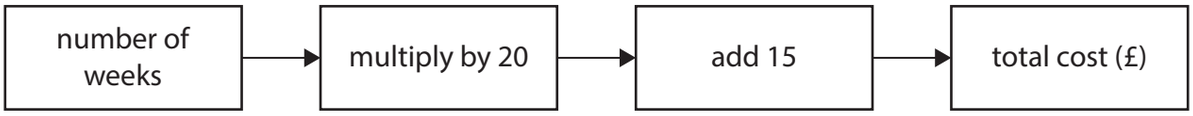
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Pearson

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Russell wants to hire a cement mixer.
He uses this formula to work out the total cost.



Russell wants to hire the mixer for 4 weeks.
He thinks the total cost will be less than £100

Is Russell correct?
Show why you think this.

(3)

Step 1: Multiply number of weeks by 20:

- 4 weeks × £20 = £80 [M1]

Step 2: Add 15 to find total cost:

- Total cost = £80 + £15 = £95 [M1]

Step 3: Compare with Russell's statement (less than £100):

- £95 < £100, so the total cost is indeed less than £100.

Conclusion:

Yes, Russell is correct. [A1]

Yes [A1]

(Total for Question 1 is 3 marks)

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2

(a) Rob throws a fair coin once.
Write down the probability the coin lands on Heads.

(1)

Fair coin has 2 equal outcomes: Heads or Tails:

- *Probability(Heads) = 1/2 (or 0.5 / 50%) [A1]*

1/2 [A1]

(b) Write the following numbers in order.
Start with the smallest number.

(1)

0.4 0.31 0.371 0.130 0.38

Compare to 3 decimal places: 0.130 < 0.310 < 0.371 < 0.380 < 0.400

0.130, 0.31, 0.371, 0.38, 0.4 [B1]

Here are two mixed numbers $7\frac{3}{10}$ and $7\frac{2}{5}$

(c) Which of these two mixed numbers is greater?
Show why you think this.

(2)

- *Convert 2/5 to tenths: 2/5 = (2 × 2) / (5 × 2) = 4/10 [M1]*
- *Compare: 7 4/10 > 7 3/10, so 7 2/5 is greater. [A1]*

(Or decimal comparison: 7.4 > 7.3)

7 2/5 [A1]

(Total for Question 2 is 4 marks)

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3 Mo has 5 hens in his garden.

12 kg of hen food costs £10.50

Mo knows 1 hen needs 150 g of this food each day.

Work out the cost of this food for the 5 hens for one day.
Give your answer to the nearest penny.

(5)

Step 1: Calculate total daily food for 5 hens:

- *5 hens × 150 g = 750 g of food each day [M1]*

Step 2: Convert 12 kg bag into grams:

- *12 kg = 12 × 1000 = 12 000 g [M1]*

Step 3: Calculate how many days the 12 kg bag lasts:

- *12 000 g ÷ 750 g = 16 days [M1]*

Step 4: Calculate the daily food cost in pence:

- *Total bag cost = £10.50 = 1050 pence*
- *Cost per day = 1050 p ÷ 16 days = 65.625 pence [M1]*

Step 5: Round to the nearest penny:

- *65.625 p rounds up to 66 p [A1]*

(Continued on next page for alternative check and answer -->)

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Alternative Method Check (cost per kg):

- Cost per kg = £10.50 ÷ 12 kg = £0.875 per kg
- Daily food in kg = 750 g ÷ 1000 = 0.75 kg
- Daily cost = 0.75 kg × £0.875 = £0.65625 = 65.625 p
- **Nearest penny = 66 p (Confirmed)**

66 [A1] p

(Total for Question 3 is 5 marks)

4 Nazrin uses an app to record her sleep quality score.

She has this information for last week.

	Mon	Tues	Wed	Thu	Fri	Sat	Sun
sleep quality score	88	80	69	73	75	81	85

Nazrin thinks the range of her sleep quality scores for last week is less than 20

(a) Is Nazrin correct?
Show why you think this.

(2)

Step 1: Identify the highest and lowest scores from the table:

- *Highest = 88 (Monday); Lowest = 69 (Wednesday)*

Step 2: Calculate the range (Highest – Lowest):

- **Range = 88 – 69 = 19 [M1]**

Step 3: Compare range with Nazrin's statement (less than 20):

- *19 < 20, so the range is indeed less than 20.*

Conclusion:

Yes, Nazrin is correct. [A1]

Yes *[A1]*

(b) Show a check of your answer.

(1)

Reverse calculation check of the range:

- **69 + 19 = 88 [B1]**

(Re-creates the highest score of 88)

69 + 19 = 88 *[B1]*

(Total for Question 4 is 3 marks)

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5 Here is some information about the test scores of forty students.

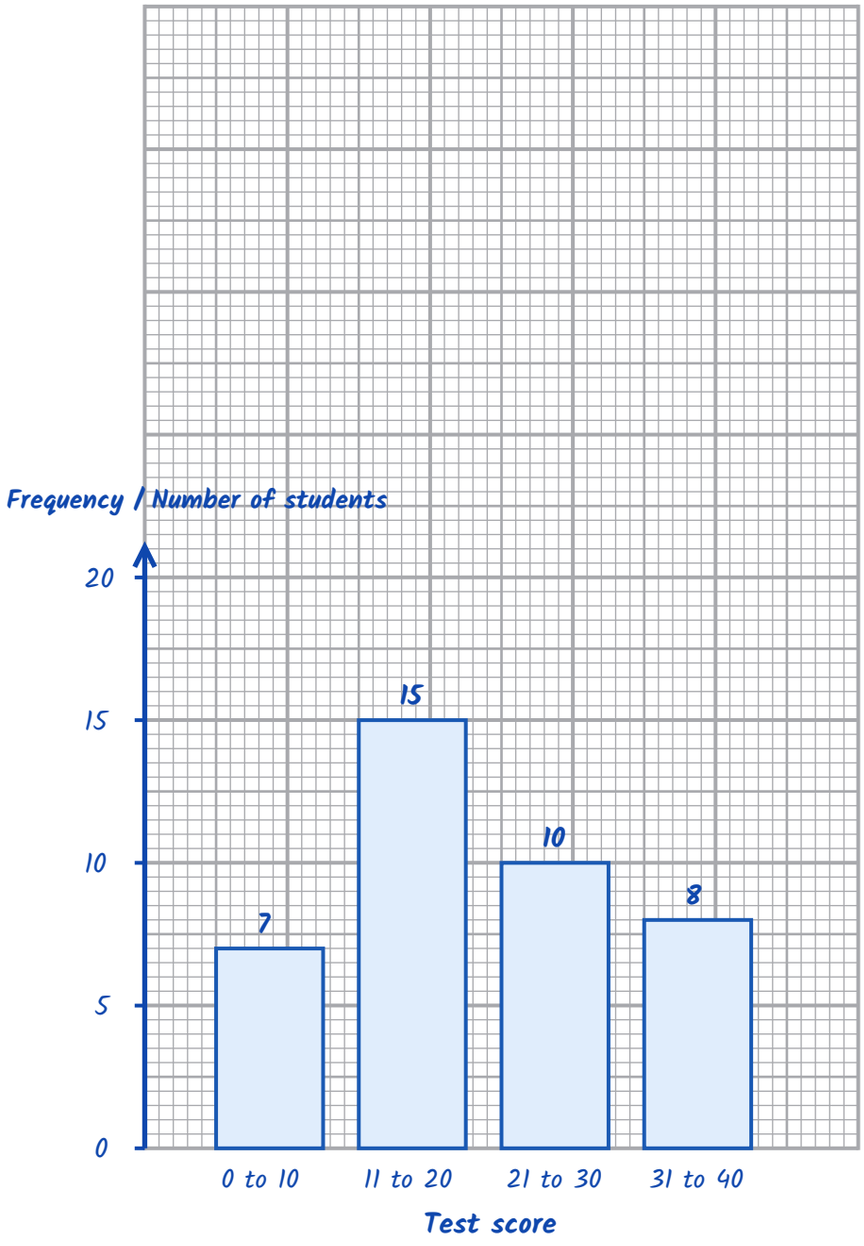
test score	0 to 10	11 to 20	21 to 30	31 to 40
number of students	7	15	10	8

On the grid below draw a bar chart for this information.

(3)

Student Test Scores Bar Chart

[M1, M1, A1]



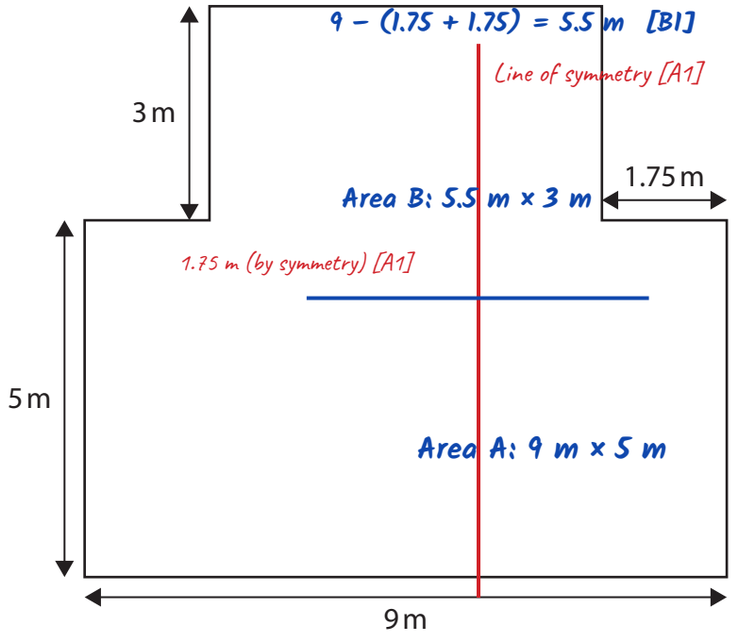
(Total for Question 5 is 3 marks)

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6 Steph wants to cover part of her garden with paving blocks.
The diagram shows the part of the garden she will cover with blocks.

The shape of this part of the garden has one line of symmetry.

All corners are right angles.



Paving blocks are sold in packs.
Each pack covers 10 m^2

Blocks can be cut and joined.

Work out the number of packs Steph needs to cover this part of her garden.
You **must** show your working.

(6)

Step 1: Calculate the width of the top middle section using symmetry:

- Width = 9 m – (1.75 m + 1.75 m) = 9 – 3.5 = 5.5 m [A1, B1]

Step 2: Calculate the area of the lower main rectangle (Area A):

- Area A = 9 m × 5 m = 45 m² [B1]

Step 3: Calculate the area of the upper middle rectangle (Area B):

- Area B = 5.5 m × 3 m = 16.5 m² [C1]

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Step 4: Calculate total garden area to be covered:

- **Total area = Area A + Area B = 45 + 16.5 = 61.5 m² [D1]**

Step 5: Calculate number of packs needed (each pack covers 10 m²):

- **Packs needed = 61.5 ÷ 10 = 6.15 packs [E1]**

Step 6: Round up to full packs (blocks can be cut and joined):

- **Steph must buy 7 packs (6 packs = 60 m², which is not enough). [F1]**

Alternative Bounding Box Check:

- *Bounding box = 9 × (5 + 3) = 9 × 8 = 72 m²*
- *Two corner cutouts = 2 × (1.75 × 3) = 10.5 m²*
- *Garden area = 72 − 10.5 = 61.5 m²; 61.5 ÷ 10 = 6.15 --> 7 packs (Confirmed)*

7 *[F1]*

(Total for Question 6 is 6 marks)

7 Kofi works in a snack kiosk.

Kofi has the following information about 90 items he sold yesterday.

item	number of sales
ice creams	34
hot drinks	12
lollies	44

[A1]

Kofi wants to draw a pie chart to display this information outside the kiosk.

Work out the size of the angle needed to represent the sales of lollies.

(3)

Step 1: Calculate the number of lollies sold:

- Lollies = 90 − (34 + 12) = 90 − 46 = 44 lollies [M1]

Step 2: Calculate degrees per item on the pie chart (360° total):

- Degrees per item = 360° ÷ 90 items = 4° per item [M1]

Step 3: Calculate the angle representing lollies:

- Angle for lollies = 44 items × 4° = 176° [M1, A1]

Verification check of full pie chart angles:

- Ice creams: 34 × 4° = 136°
- Hot drinks: 12 × 4° = 48°
- Sum = 136° + 48° + 176° = 360° (Confirmed)

176 [A1] °

(Total for Question 7 is 3 marks)

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8 Dionne sells comics online.

On Tuesday there were 56 orders from customers.

$\frac{3}{8}$ of the customers used vouchers when they paid for their orders.

Dionne thinks 23 of the customers on Tuesday used a voucher.

Is Dionne correct?
Show why you think this.

(3)

Step 1: Calculate 1/8 of the 56 customer orders:

- 56 ÷ 8 = 7 customers [M1]

Step 2: Calculate 3/8 of the 56 customer orders:

- 3 × 7 = 21 customers used vouchers [M1]

Step 3: Compare with Dionne's statement (23 customers):

- 21 ≠ 23 (The actual number is 21, not 23).

Conclusion:

No, Dionne is not correct. [A1]

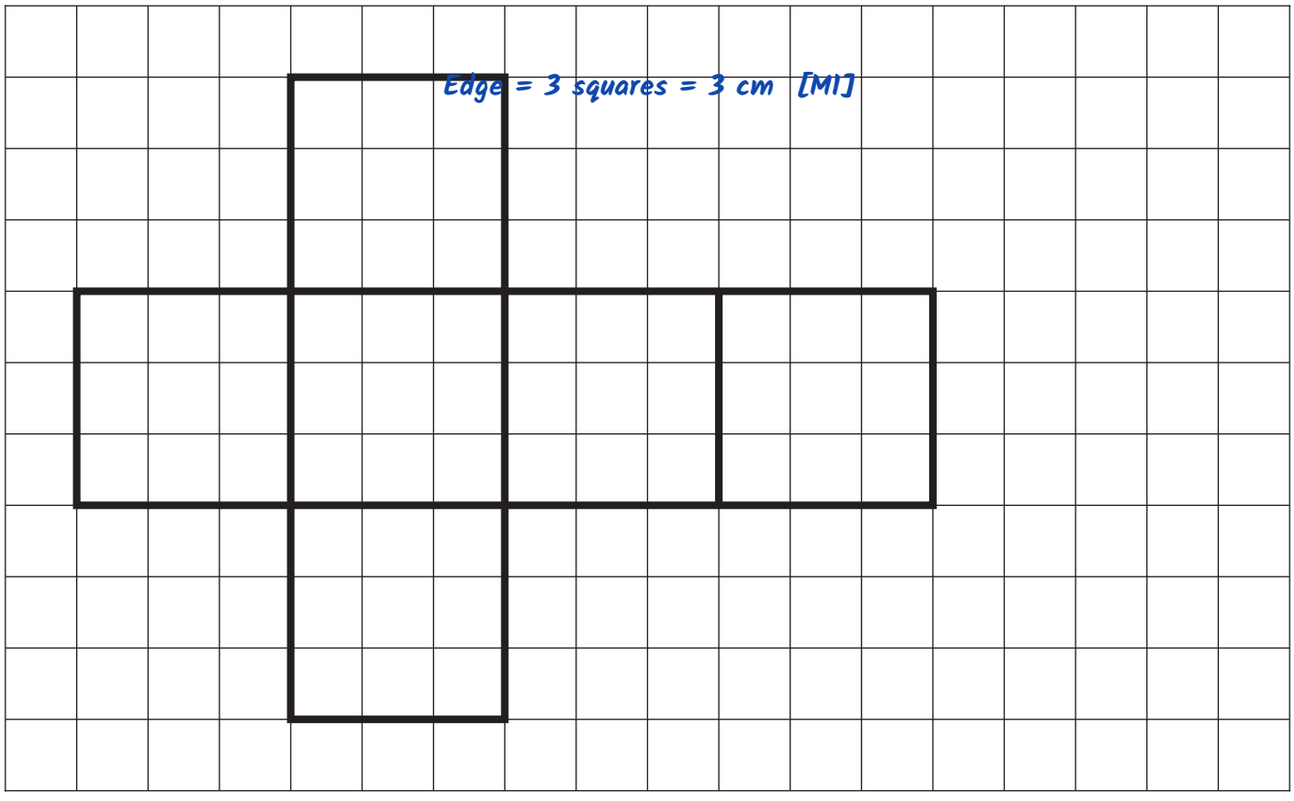
Alternative Fraction Check:

- 23 / 56 ≈ 0.411, but 3 / 8 = 0.375 (Does not match)

No [A1]

(Total for Question 8 is 3 marks)

9 Here is an accurate net of a cube drawn on a centimetre grid.



Work out the volume of the cube.
Remember to state your units.

(4)

Step 1: Find the edge length of the cube from the centimetre grid net:

- *1 square = 1 cm; each face is 3 squares wide: edge length = 3 cm [M1]*

Step 2: Formula for the volume of a cube:

- *Volume = length × width × height = edge³*
- **Volume = 3 cm × 3 cm × 3 cm = 27 cm³ [M1, A1]**

Units required: cm³ [B1]

27 cm³ *[A1, B1]*

(Total for Question 9 is 4 marks)

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10 Charlie is planning a new room in a hotel.
He has a diagram of the room drawn on a grid.

Charlie needs to show a space for the wardrobe.

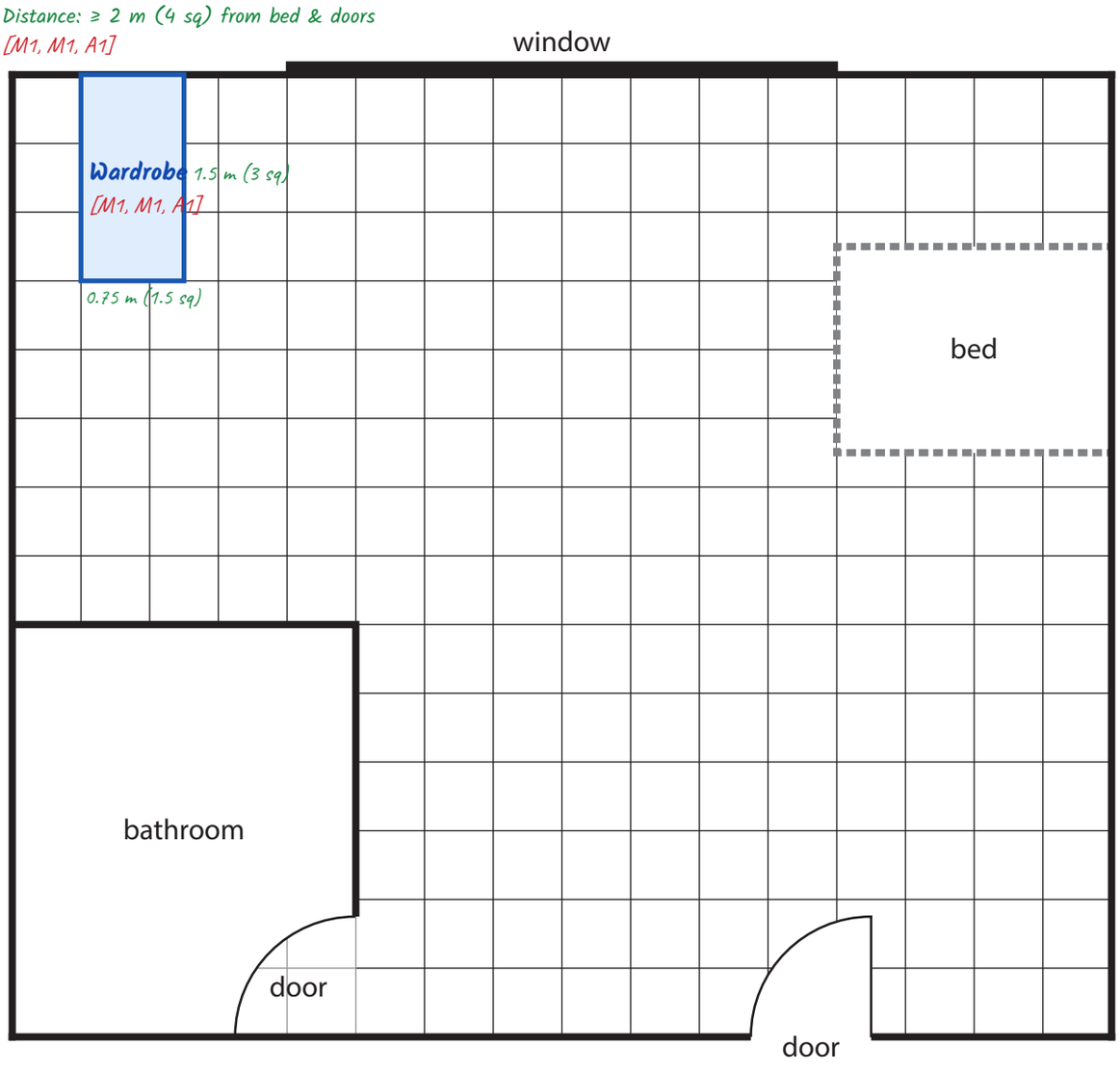
The space for the wardrobe must

- be rectangular in shape 1.5 m by 0.75 m
- have the longest side against a wall
- be at least 2 m away from the doors
- be at least 2 m away from the bed.

Draw a space for the wardrobe on the grid below.

(3)

Scale: 1.5 m = 3 sq (on wall); 0.75 m = 1.5 sq
Distance: ≥ 2 m (4 sq) from bed & doors
[M1, M1, A1]



Key 1 cm on the grid represents 50 cm in the hotel room
wall

(Total for Question 10 is 3 marks)

11

(a) Write seventy percent in figures and symbols.

(1)

70% [A1]

Ahmed works in a museum.
Last year there were six hundred and eighty thousand visitors to the museum.

This year the museum was closed for three months.
Ahmed expects the number of visitors to be 25% less this year than last year.

(b) Work out the number of visitors Ahmed expects this year.

(4)

Step 1: Write last year's visitor number in figures:

- **Six hundred and eighty thousand = 680 000 visitors [B1]**

Step 2: Calculate the 25% reduction in visitors (1/4 of total):

- 25% = 1/4
- **680 000 ÷ 4 = 170 000 fewer visitors [M1]**

Step 3: Subtract reduction from last year's total:

- Expected visitors = 680 000 − 170 000
- **Expected visitors = 510 000 visitors [M1, A1]**

(Continued on next page for alternative multiplier method -->)

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Alternative Multiplier Method Check:

- 100% − 25% = 75% = 0.75 of last year's visitors
- Expected visitors = 680 000 × 0.75 = 510 000 (Confirmed)

510 000 [A1]

(Total for Question 11 is 5 marks)

TOTAL FOR SECTION B IS 42 MARKS
TOTAL FOR PAPER IS 56 MARKS

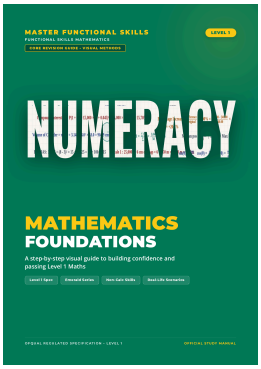
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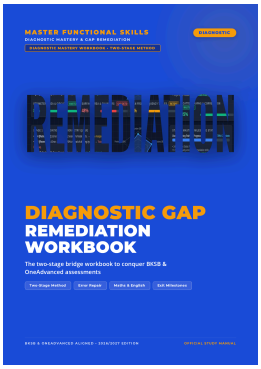


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